2.3.7 Extensions of Mendelian Genetics

Apart from monohybrid, dihybrid and trihybrid crosses, there are exceptions to Mendelian principles, i.e. the occurrence of different phenotypic ratios. The more complex patterns of inheritance are the extensions of Mendelian Genetics. There are examples where phenotype of the organism is the result of the interactions among genes.

Gene interaction – A single phenotype is controlled by more than one set of genes, each of which has two or more alleles. This phenomenon is called Gene Interaction. Many characteristics of the organism including structural and chemical which constitute the phenotype are the result of interaction between two or more genes.

9TL2WA

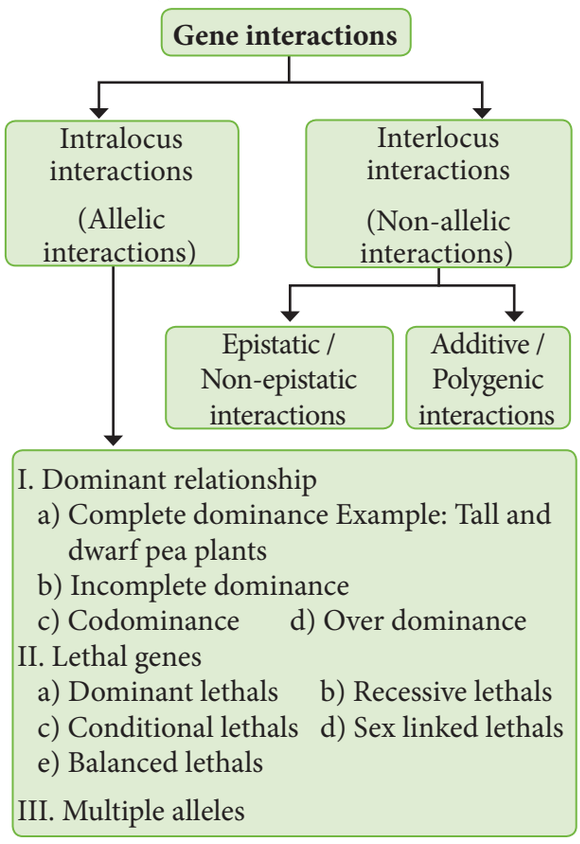

Figure 2.13: Gene Interaction

Mendelian experiments prove that a single gene controls one character. But in the post Mendelian findings, various exception have been noticed, in which different types of interactions are possible between the genes. This gene interaction concept was introduced and explained by W. Bateson. This concept is otherwise known as Factor hypothesis or Bateson's factor hypothesis. According to Bateson's factor hypothesis, the gene interactions can be classified as

- Intragenic gene interactions or Intra allelic or allelic interactions
- Intergenic gene interactions or inter allelic or non-allelic interactions

2.4 Intragenic interactions

Interactions take place between the alleles of the same gene i.e., alleles at the same locus is called intragenic or intralocus gene interaction. It includes the following:
1) Incomplete dominance (2) Codominance (3) Multiple alleles (4) Pleiotropic genes are common examples for intragenic interaction.

2.4.1. Incomplete dominance – No blending of genes

The German Botanist Carl Correns's (1905) Experiment - In 4 O' clock plant, *Mirabilis jalapa* when the pure breeding homozygous red (R^1R^1) parent is crossed with homozygous white (R^2R^2), the phenotype of the F_1 hybrid is heterozygous pink (R^1R^2). The F_1 heterozygous phenotype differs from both the parental homozygous phenotype. This cross did not exhibit the character of the dominant parent but an intermediate colour pink. When one allele is not completely dominant to another allele it shows incomplete dominance. Such allelic interaction is known as incomplete dominance. F_1 generation produces intermediate phenotype pink coloured flower. When pink coloured plants of F_1 generation were interbred in F_2 both phenotypic and genotypic ratios were found to be identical as 1 : 2 : 1(1 red : 2 pink : 1 white). Genotypic ratio is 1 R^1R^1 : 2 R^1R^2 : 1 R^2R^2.From this we conclude that the alleles themselves remain discrete and unaltered proving the Mendel's Law of Segregation. The phenotypic

TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_033-052 CH 02.indd 40 3/14/2020 10:30:53 AM